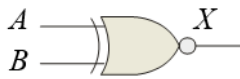


Digital Logic Circuits (Spring 2026)

Quiz #2

- The 2's complement of 1000 is 1000.
- The fractional binary number 0.11 has a decimal value of 0.75.

- The symbol  is XNOR gate.

- A 2-input gate produces a HIGH output only when the inputs agree. This type of gate is XNOR gate.

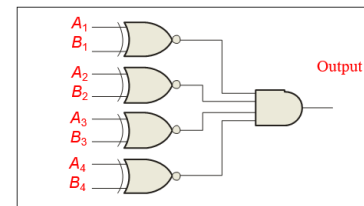
- The minimum expression that can be read from the Karnaugh map shown in the right is _____.

$$X = \bar{B}$$

	$\bar{C}$	C
$\bar{A}B$	1	1
$\bar{A}\bar{B}$		
AB		
$A\bar{B}$	1	1

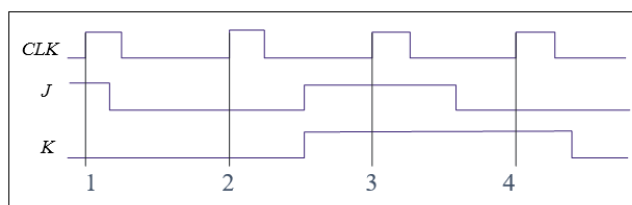
- Assume an AOI (AND-OR-Invert) expression is $\overline{AB + CD}$. The equivalent POS (product-of-sum) expression is $(\bar{A} + \bar{B})(\bar{C} + \bar{D})$.

- In the right figure, the output will be LOW if **c**
 - $A < B$
 - $A > B$
 - both a and b are correct
 - $A = B$



- The output of a D latch (shown below) will not change if **b**
 - the output is LOW
 - Enable is not active
 - D is LOW
 - all of the above

- Assume the output is initially HIGH on a leading edge triggered J-K flip flop. For the inputs shown below, the output will go from HIGH to LOW on which clock pulse? **c**
 - 1
 - 2
 - 3
 - 4



- The circuit illustrated right is a **d**
 - astable multivibrator
 - data storage device
 - frequency multiplier
 - frequency divider

